KU LEUVEN

FACULTY OF SCIENCE

The State of Comon's Conjecture

Lael JOHN

Supervisor: Prof. N. Vannieuwenhoven
KU Leuven

Co-supervisor: Dr. D. Taufer
KU Leuven

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Contents

List of Notation

General Objects

$[n]$	The set $\{1, 2, \ldots, n\}$
$\mathbb{F}$	A general field, in this thesis either $\mathbb{R}$ or $\mathbb{C}$ unless specified
$\mathbb{F}[x_1, \ldots, x_n]_d$	Subring of polynomial ring in n variables over $\mathbb{F}$, consisting of polynomials of fixed degree d
σ	A permutation of a finite collection of elements (the length of the collection is specified in context)

Vector Spaces

$\hat{V}_i$	The tensor product $V_1 \otimes \cdots \otimes V_{i-1} \otimes \left(\bigoplus_{\lambda \in \Lambda} \mathbb{F}\right) \otimes V_{i+1} \otimes \cdots \otimes V_d$
$\langle v_i \rangle_{i=1}^n$	The linear span of n vectors
$S^d(\mathbb{F}^n)$	Symmetric tensors (forms) of order d over $\mathbb{F}^n$
V_i	A vector space over $\mathbb{F}$ having dimension n_i
V_i^*	The dual vector space to V_i
$V_{\hat{i}}$	The tensor product of vector spaces excluding V_i, defined as $V_1 \otimes \cdots \otimes V_{i-1} \otimes V_{i+1} \otimes \cdots \otimes V_d$

Vectors

$\mathbf{1}_n$	The vector of all 1's lying in $\mathbb{F}^n$
α	A vector of non-negative integers $(a_1, \ldots, a_n)$
$\mathbf{x}^\alpha$	$\prod_{i=1}^n x_i^{a_i}$ where n is taken from context
$\mathbf{x}$	A vector of indeterminates $(x_1, \ldots, x_n)$
$e_{i,j}$	The j-th standard ordered basis vector in V_i. The i may not be dropped if it is clear from context
$v(k)$	The kth entry in a vector v
v_i^k	The k-th vector lying in V_i

Tensors

$\bar{\mathcal{M}}$	The tensor constructed by taking elements of $\mathcal{M}$ as slices
$\mathcal{M}$	A finite family of tensors
$T(k_1, \dots, k_d)$	The element located at the position $(k_1, \dots, k_d)$ in a tensor T
T_j^i	The j-th slice along the i-th mode of a tensor T (see Definition 4.1.1)
$T^{(i)}$	The flattening of a tensor T along the ith mode (see Definition 4.1.2)
$T^{\emptyset}$	The unravelling of a tensor T into a vector, with elements indexed by $\times_{i=1}^{d}[n_i]$ in lexicographic order (see remarks after Definition 4.1.2)
$T^{\otimes d}$	The tensor product of T with itself d times, $\underbrace{T \otimes T \otimes \cdots \otimes T}_{d \text{ times}}$

General Functions

$\pi_N^i(T)$	The mapping denoting a projection down to the vector space $\mathbb{F}^N$ along the i-th mode of a tensor (see Definition 4.2.3)
$Adj(T, \mathcal{M}_1, \dots, \mathcal{M}_d)$	The general adjoining of collections of slices $\mathcal{M}_i$ along all i modes of a tensor T
$Adj^i(T, \bar{\mathcal{M}})$	The adjoining of $\mathcal{M}$ as slices along the i-th mode of a tensor T (see Definition 4.2.2)
$C_n(T)$	The n-th clone of a tensor T
$F^{(i)}(T)$	The mapping denoting the flattening of a tensor T along the i-th mode
$Mod^i(T, \bar{\mathcal{M}})$	The mapping generating a family of tensors by slicing T with elements from $\mathcal{M}$ along the ith mode (see Definition 4.2.6)
$pad^i(A, T)$	The padding of a lower order tensor A into the same format as T by inserting 0's along the ith mode (see Definition 4.2.4)
$Pad_j^i(A, T)$	The mapping taking a lower order tensor A into the same format as a higher order tensor T, now seen as the only non-zero j-th slice along the ith mode (see Definition 4.2.5)
$S_j^i(T)$	The mapping that selects the j-th slice along the i-th mode of a tensor T
$SAdj(T, \mathcal{M})$	The symmetric adjoining operation defined for a symmetric tensor T and a collection of symmetric slices $\mathcal{M}$, $Adj(T, \mathcal{M}, \dots, \mathcal{M})$

$T \bmod (\mathcal{M}_1, \ldots, \mathcal{M}_d)$	Family generated by slicing T along each i-th mode with slices from $\mathcal{M}_i$

Scalar-Valued Functions

Rk(T)	The rank of a tensor T
SRk(T)	The symmetric rank of a symmetric tensor T

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Chapter 1

Introduction

The title of this thesis centers around an open question about tensors. These are mathematical tools, used to describe multiple linear relationships that could exist between mathematical objects. In their doing so, tensors allow for the description of a *multi*-linear algebra describing how the objects relate to each other. The focus of this thesis is threefold: on how tensors can be built, where they live, and how they can be broken down.

The current state of Comon's conjecture (an open problem for the last 20 years) is not easily described. While mathematicians have chipped away at the number of cases where it remains valid, an answer in full generality over $\mathbb{R}$ or $\mathbb{C}$ remains unknown. The hope is that at least some of the work done in recent years can be highlighted as useful, paving the way towards a definitive counterexample to- or proof of the conjecture.

This question about tensors merits study because of the powerful nature of tensors as descriptors of multilinear relationships. One of the more important problems that has come to light in recent years is the computational complexity of matrix multiplication ([18]), and tensors and their rank help to quantify exactly how efficient the algorithms for matrix multiplication are. Tensors also find their use in quantum mechanics (describing the entanglement of particles), general relativity (as tools to describe the curvature of space-time [9]) , chemometrics, psychometrics (representing combinations of chemicals or traits that are interrelated [28]), in data science and computer vision [20] and signal processing (the source selection problem [18]) — to name some

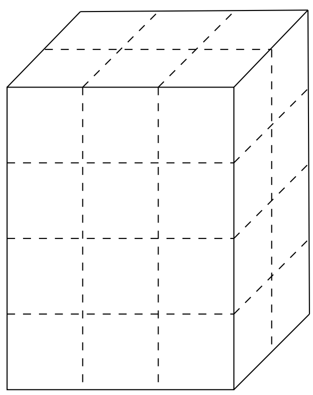

Figure 1: Example of a tensor $T \in \mathbb{F}^4 \otimes \mathbb{F}^3 \otimes \mathbb{F}^2$, represented in the form of a 3-way array

more examples. In each of these cases, knowing how important symmetry is as a feature of the relationship would play a vital role in analysing procedures and algorithms that take for granted that symmetry as the more ”natural” state of affairs. This forms the guiding idea behind the ideas and techniques explored during this thesis.

Comon’s Conjecture

At its heart, Comon’s Conjecture centers around two central themes: *rank* and *symmetry* in tensors. Rank (see definition 2.2.3) describes how a tensor can be decomposed into simpler constituent tensors, and symmetry (see definition 2.3.2) describes how tensors behave under permutation actions.

The conjecture was first stated openly in July 2008 at ”Geometry and Representation Theory of Tensors for Computer Science, Statistics and Other Areas” - a workshop organised at the American Institute for Mathematics [19] in the group focusing on ”Multilinear Techniques in Data Analysis and Signal Processing”. The conjecture as stated in the report is as follows

Conjecture (Comon, (Oeding 2008 [19]))**.** *Do the rank and symmetric rank of T always coincide, where T is a symmetric, real or complex tensor?*

This report also covered work done previously by Comon et al. [13] where they also showed that the conjecture held for situations when the symmetric rank was at least as large as the dimension of the underlying vector space, or when the order (see definition 2.1.1) of T was sufficiently large. Since 2008, the mathematical community has proved the conjecture true under various assumptions (see chapter 3).

In 2018, Shitov [22] proposed the construction of a counterexample living in $(\mathbb{C}^{800})^{\otimes 3}$ with symmetric rank at least 904, and a rank of at most 903. In 2024 however, an erratum published [14] pointed out an error in the original paper, rendering the conjecture still open, at least for the complex case. In the intervening time, a paper published by Wang and Seigal [27] demonstrated the construction of a real, order 6, symmetric tensor having real rank 30913 and symmetric rank 30914, thus disproving the conjecture for the real case. This paper built on a method of construction developed in a preprint from Shitov in 2020 [23]. These ideas have also been developed further in subsequent preprints - [24] and [25], again from Shitov.

The following chapters and sections highlight important terminology required to state and describe the setting of Comon’s conjecture, highlight the work done to prove the conjecture in the past, and they finally describe the major constructions required to construct the real counterexample.

Chapter 2 This chapter covers preliminary terminology and definitions required to understand the history of attempts to prove Comon’s conjecture. It also serves to show the different manners of viewing tensors.

Chapter 3 This chapter goes over the literature that currently exists since the statement of the problem in 2008. The papers discussed highlight the valid attempts at proving Comon’s conjecture over $\mathbb{R}$ and $\mathbb{C}$

Chapter 4 This chapter focuses on understanding Shitov's methodology and approach to constructing a counterexample to Comon's conjecture. The culmination of this chapter is an examination of the paper from Wang and Seigal [27], a published real-valued counterexample to the conjecture.

Chapter 5 This chapter highlights the gaps present in the literature so far, illustrates problems with the current approach to constructing counterexamples, and seeks to highlight potential future avenues to solving this problem once and for all.

The personal contributions made in the process of writing thesis are in the compilation of existing literature surrounding Comon's conjecture, establishing lemmas required for the constructions made by Shitov, and (wherever possible) either generalising the arguments made by Wang and Seigal or showing why a generalisation is not possible.

Chapter 2

Preliminaries Regarding Tensors

2.1 Tensors in an Algebraic Setting

The most abstract way of thinking about tensors is when describing them as elements of a tensor product of modules. Consider A, B and P as being R-modules, where R is a ring. Then any bi-linear mapping, $t : A \times B \to P$, is described by acting on elements in the following manner:

$$t(\lambda a, b) = t(a, \lambda b) = \lambda t(a, b), a \in A, b \in B, \lambda \in R, t(a, b) \in P$$

$$\begin{tikzcd}
A \times B \arrow[rr, "t \text{ bilinear}=\tilde{t}\circ\varphi"] \arrow[dr, "\varphi"'] & & P \\
& A \otimes_R B \arrow[ur, "\exists ! \tilde{t}"'] &
\end{tikzcd}$$

Figure 2: Illustrating a tensor product of R-modules

Such a bilinear (or even multilinear mapping) between R-modules can be represented by an element lying in their tensor product, such that the following diagram of modules commutes [26] (illustrated specifically for the case mentioned above in fig. 2). This act of representing any such bilinear mapping via an element of the tensor product is commonly referred to as a *universal property* in algebra.

While such an abstract description is useful when showing how any general elements lying in the specified spaces can be combined into a tensor, it does not specify details required for actually computing how this tensor can be represented.

A more useful approach to understanding tensors is to begin from linear algebra. As matrices represent linear transformations between vector spaces V and W, so d-way arrays can be used to represent multilinear relationships between vector spaces $V_1, V_2 \dots V_d$, where each V_i has a dimension n_i. Diving deeper into this analogy, matrices of the form $n_i \times n_j$ behave as either bilinear maps from the product $V_i \times V_j \to \mathbb{F}$, or as linear maps $V_j \to V_i^*$. This hybrid nature of matrices is a reflection of the underlying fact that matrices are the same as order 2 tensors. Since computing elements of the matrix A_{ij} depends on a choice of bases for both vector spaces V and W, an analogous choice of bases must be made to represent the order d tensor $T \in V_1 \otimes V_2 \otimes \dots \otimes V_d$ as a d-way array.

For the purposes of this thesis, unless specified, each vector space will be considered over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. Here each vector space V_i is equipped with its standard ordered basis $\{e_{i,j}\}_{j=1}^{n_i}$ where $e_{i,j}(k) = \delta_{j,k}$. Each different choice of basis changes how the tensor T is represented as a d-way array.

Definition 2.1.1 (Order of a Tensor)**.** A tensor $T \in V_1 \otimes \cdots \otimes V_d$ is said to be an **order** d **tensor** because it lies in the tensor product of d vector spaces.

Consider $T := v_1 \otimes \cdots \otimes v_d$, a particular order-$d$ tensor, with each component vector $v_i \in V_i$, with components $v_i(j)$ according to the standard ordered basis of the corresponding vector space. Then the elements of a d-way array that corresponds to T can be given in the following manner:

$$T(k_1, k_2, \ldots, k_d) = \prod_{i=1}^{d} v_i(k_i),$$

where $0 \leq k_i \leq n_i$— the dimension of each corresponding vector space V_i.

In fact, for a general order d tensor $T \in V_1 \otimes \cdots \otimes V_d$, one can form a similar d-way array to represent it, by using the fact of rank decompositions (see definition 2.2.2)

Remark. For a tensor T of order d, the *modes* of T refer to the d distinct elements that make up the tuples indexing elements in the d-way array that represents T.

Example 1. Consider

$$T := (a, b, c, d) \otimes (x, y, z) \otimes (p, q) \in \mathbb{F}^4 \otimes \mathbb{F}^3 \otimes \mathbb{F}^2$$

. Then T is an order 3 tensor, with 3 modes, since each entry in the 3-way array representing T is indexed by a tuple $(i_1, i_2, i_3) \in [4] \times [3] \times [2]$

$T := v_1 \otimes \cdots \otimes v_d \in V_1 \otimes \cdots \otimes V_d$ can be viewed in the following manner:

1. As a multilinear mapping between $V_1^* \times V_2^* \times \cdots \times V_d^* \to \mathbb{F}$ (extending the analogue of looking at matrices as bilinear forms)

$$(f_1, f_2 \ldots f_d) \mapsto \prod_{i=1}^{d} f_i(v_i).$$

2. As a linear mapping from $V_i^* \to V_1 \otimes \cdots \otimes V_{i-1} \otimes V_{i+1} \otimes \cdots \otimes V_d =: V_{\hat{i}}$ (moving from a space of order-d tensors to one of order-$d-1$)

$$f_j \mapsto f_j(v_i)(v_1 \otimes \cdots \otimes v_{i-1} \otimes v_{i+1} \otimes \cdots \otimes v_d). \tag{2.1}$$

This viewpoint can be seen in section 4.1.3, slicing the tensor along the ith mode.

3. As the dual of the defined linear map (see eqn(2.1)), now going from $V_{\hat{i}}^* \to V_i$ (moving now from a tensor of order order-d to a vector collinear to v_i)

$$(f_j)_{j \neq i} \mapsto \left(\prod_{j \neq i} f_j(v_j) \right) v_i.$$

Each of these above viewpoints also corresponds to an appropriate output to the transformation of the representative d-way array chosen: either to a scalar, a $d-1$-way array or a vector respectively [18]. In fact scalars and vectors can be defined as tensors of orders 0 and 1 respectively.

2.2 Rank

The first half of Comon's conjecture involves figuring out how tensors can be built, since the following example shows that there are tensors that cannot be constructed as simply the tensor product of vectors. As an analogy, not all vectors in a vector space can be written as only scaled basis vectors.

Example 2. When dealing with 3×3 matrices $\in \mathbb{F}^3 \otimes \mathbb{F}^3$, consider $e_1 \otimes e_1 + e_2 \otimes e_2$. Such a matrix cannot be written as a simple tensor product of two vectors $v_1 \otimes v_2$, where $v_1, v_2 \in \mathbb{F}^3$.

Definition 2.2.1 (Elementary Tensor)**.** Consider an order d tensor, T with the specific form $T := v_1 \otimes \cdots \otimes v_d \in V_1 \otimes \cdots \otimes V_d$. Since T can be written explicitly as the tensor product of d vectors, it is called an **elementary tensor**. Equivalent names for such tensors in the literature reviewed are *rank one* tensors and *decomposable* tensors.

Using these elementary tensors as building blocks provides a way to describe the construction of more general tensors.

Definition 2.2.2 (Tensor Decomposition, (Zhang 2016 [28]))**.** Consider a general order d tensor $T \in V_1 \otimes \cdots \otimes V_d$. A **decomposition** of T consists of writing T as a sum of elementary tensors

$$T = \sum_{k=1}^{s} v_1^k \otimes \cdots \otimes v_d^k,$$

with each of the $v_i^k \in V_i$.

Remark. Here note that there exist multiple names for describing the decomposition of a tensor into the sum of elementary tensors - polyadic forms, canonical decomposition (CANDECOMP), paralell factors (PARAFAC) etc. The above definition agrees with what is known in the literature as CP-decomposition (a combination of CANDECOMP/PARAFAC). [16]

For any given tensor T now, there can exist multiple decompositions, not just depending on the choice of basis vectors for each V_i. Two questions naturally follow this definition

Q1. For a given order d tensor T, does there exist a shortest decomposition?

Q2. If such a shortest decomposition exists, is it unique (apart from the choice of basis vectors)?

The answer to the first question can be given in terms of the following definition.

Definition 2.2.3 (Tensor Rank)**.** The **rank** of a tensor T, denoted **Rk**(T), is the minimal length of any possible decomposition of T.

Tensor rank always exists, because tensors can always be written in terms of elementary tensors. Computing the rank of a given tensor is not an easy problem, but there now exist multiple approaches in the literature that rely on both the algebraic properties of tensors and associated transformations, and the geometric properties of the varieties on which these tensors lie. Tensor rank also remains invariant of component tensors involved in a decomposition, and is one of the two key ingredients required to state Comon's Conjecture as specified in chapter 1. Thus Q1 as given above has an affirmative answer, but Q2 does not - there can exist multiple rank decompositions.

2.3 Symmetric Tensors

The other half of Comon's Conjecture revolves around tensors that exhibit the properties of symmetry. These become valuable in applications simply because the symmetry means that they require "less" information to be constructed. What makes a tensor symmetric is the next logical question to tackle, and makes it necessary to define how permutations act on a given tensor.

Given $\sigma \in S_d$ (the symmetry group $\{1, 2, \ldots, d\}$), one can define the permutation action of σ on a tensor T as follows:

Definition 2.3.1 (Permutation of a Tensor)**.** Let $T \in V_1 \otimes \cdots \otimes V_d$, where $V_i = V_j,\ \forall i, j$. Then $\sigma \in S_d$ acts on T (represented as a d-way array) in the following manner

$$\sigma \cdot T(k_1, k_2, \ldots k_d) = T(\sigma(k_1), \sigma(k_2), \ldots, \sigma(k_d))$$

Note here that all the vector spaces are the same, to keep this permutation mapping well defined. This fact also allows for the use of notation $T \in V^{\otimes d}$.

The definition of a symmetric tensor follows almost immediately

Definition 2.3.2 (Symmetric Tensor [13])**.** Consider a tensor $T \in V^{\otimes d}$. T is said to be **symmetric** if the following equality holds

$$T = \frac{1}{d!} \sum_{\sigma \in S_d} \sigma \cdot T$$

The requirement of $T = \sigma \cdot T$ for all $\sigma \in S_d$ follows as a consequence of this definition.

Since every tensor can be expressed as a sum of elementary tensors (see definition 2.2.2), **Rk**(T) is defined for symmetric tensors T as well. However, one can define *elementary symmetric tensors* as tensors in $V^{\otimes d}$ of the form $v^{\otimes d}$ where $v \in V$. Establishing this definition allows for symmetric tensors to be written as a sum of these elementary symmetric tensors, resulting in the definition of **symmetric decompositions** and the analogous notion of **symmetric rank**, denoted **SRk**(T).

Example 3. Consider the tensor

$$T = (1,0,0)\otimes(1,0,0)\otimes(1,0,0)+(0,0,1)\otimes(0,0,1)\otimes(0,0,1)+(0,1,0)\otimes(0,1,0)\otimes(0,1,0)$$

. This is definitely a symmetric decomposition, and T is a symmetric tensor.

It is clear that not every tensor can be seen as symmetric, and therefore the subset of all order-d tensors that are symmetric is denoted $S^d(\mathbb{F}^n) \subseteq (\mathbb{F}^n)^{\otimes d}$ — notation usually reserved for homogenous polynomials in n variables of fixed degree d. In fact it can be shown that these spaces, $S^d(\mathbb{F}^n)$ and $\mathbb{F}[x_1, \ldots, x_n]_d$, are isomorphic to each other (see Proposition 2.3.1). This equivalence is a powerful tool that allows for an interchange of polynomial and tensor notation (when appropriate).

Symmetric Tensors and Homogenous Polynomials of Fixed Degree

A homogenous polynomial $f \in \mathbb{F}[x_1, \ldots, x_n]_d$ is of the form

$$f(x_1, \ldots x_n) = \sum_{a_i \geq 0, \sum_i^n a_i = d} \lambda_{(a_1,\ldots,a_n)} \prod_{i=1}^{n} x_i^{a_i},$$

where all the coefficients $\lambda_{(a_1,\ldots,a_n)} \in \mathbb{F}$. The following notation is also standard in the literature: $\mathbf{x} := (x_1, x_2, \ldots x_n)$ and $\alpha := (a_1, a_2, \ldots a_n) \in (\mathbb{N} \cup \{0\})^n$. Then one has $\mathbf{x}^\alpha = \prod_{i=1}^n x_i^{a_i}$. Further $|\alpha| = \sum_i^n a_i$, so the homogenous polynomial as stated above can be rewritten as

$$f(\mathbf{x}) = \sum_{|\alpha|=d} \lambda_\alpha \mathbf{x}^\alpha.$$

The following proposition is standard in tensor decomposition literature, but is stated again for convenience.

Proposition 2.3.1. *For fixed values of $d, n > 0$, the map*

$$\Phi_d : \mathbb{F}[x_1, \ldots, x_n]_d \to S^d(\mathbb{F}^n),$$

is an isomorphism of vector spaces over $\mathbb{F}$.

Proof. Define the mapping first for all degree d monomials $\mathbf{x}^\alpha$ as

$$\mathbf{x}^\alpha := \prod_{i=1}^n x_i^{a_i} \mapsto \frac{1}{d!} \sum_{\sigma \in S_d} \sigma \cdot \left(\bigotimes_{i=1}^n e_i^{\otimes a_i} \right).$$

Thus each degree d monomial is mapped to a symmetric tensor (by construction). Here each monomial can be seen as a basis vector in the vector space $\mathbb{F}[x_1, \ldots, x_n]_d$. By linearity, this mapping is well defined, since all other homogenous degree$d$ polynomials can be constructed as linear combinations of these basis vectors. Therefore for $\lambda \in \mathbb{F}$

$$\Phi(\lambda \mathbf{x}^\alpha + \mathbf{x}^\beta) = \lambda \left(\frac{1}{d!} \sum_{\sigma \in S_d} \sigma \cdot (\bigotimes_{i=1}^n e_i^{\otimes a_i}) \right) + \left(\frac{1}{d!} \sum_{\sigma \in S_d} \sigma \cdot (\bigotimes_{i=1}^n e_i^{\otimes b_i}) \right) = \lambda \Phi(\mathbf{x}^\alpha) + \Phi(\mathbf{x}^\beta).$$

With this definition, Φ is an homomorphism of vector spaces

Further, every elementary symmetric tensor $v^{\otimes d} = (\sum_{i=1}^n v_i e_i)^{\otimes d}$ can be seen as the image of the d-th power of the linear form $(\sum_{i=1}^n v_i x_i)^d$. Since every symmetric tensor has a symmetric tensor decomposition, Φ must also be surjective.

Finally, consider two homogenous polynomials $f(\mathbf{x}) := \sum_{|\alpha|=d} f_\alpha \mathbf{x}^\alpha$ and $g(\mathbf{x}) := \sum_{|\alpha|=d} g_\alpha \mathbf{x}^\alpha$ such that $f_\alpha, g_\alpha \in \mathbb{F} \; \forall \; \alpha$, and $\Phi(f(\mathbf{x})) = \Phi(g(\mathbf{x}))$.

This is equivalent to saying

$$
\begin{aligned}
& \Phi(\sum_{|\alpha|=d} f_\alpha \mathbf{x}^\alpha) = \Phi(\sum_{|\alpha|=d} g_\alpha \mathbf{x}^\alpha) \\
\iff \quad & \sum_{|\alpha|=d} f_\alpha \Phi(\mathbf{x}^\alpha) = \sum_{|\alpha|=d} g_\alpha \Phi(\mathbf{x}^\alpha) \\
\iff \quad & \sum_{|\alpha|=d} \frac{f_\alpha}{d!} \sum_{\sigma \in S_d} \sigma \cdot \left(\bigotimes_{i=1}^{n} e_i^{\otimes a_i} \right) = \sum_{|\alpha|=d} \frac{g_\alpha}{d!} \sum_{\sigma \in S_d} \sigma \cdot \left(\bigotimes_{i=1}^{n} e_i^{\otimes a_i} \right) \\
\iff \quad & \sum_{|\alpha|=d} \frac{(f_\alpha - g_\alpha)}{d!} \sum_{\sigma \in S_d} \sigma \cdot \left(\bigotimes_{i=1}^{n} e_i^{\otimes a_i} \right) = 0 \\
\iff \quad & f_\alpha = g_\alpha \ \forall \ |\alpha| = d
\end{aligned}
$$

and this is enough to say that $f(\mathbf{x}) = g(\mathbf{x})$, showing that Φ is an injective function.Thus there exists an isomorphism between these two vector spaces. □

In fact Φ_d can be seen as a component of a larger isomorphism between graded vector spaces, $\Phi : \mathbb{F}[x_1, \ldots, x_n] \to \bigoplus_{d>0} S^d(\mathbb{F}^n)$ [13].

Chapter 3

Literature Review

Since the statement of the conjecture in 2008, there has been a substantial amount of work done to arrive at a conclusive final resolution. Some of this body of work relies on the specific geometric construction of secant varieties and how they relate to the question of computing rank and symmetric rank. This connection to geometry is not surprising because tensors as mathematical objects can also be seen as points lying in space (much like vectors and matrices can be seen as representing points, as well as algebraic objects.) This chapter goes over some of the affirmative answers to Comon's conjecture, with the final section of this chapter compiling all the conditions currently known for which the conjecture holds.

3.1 Attempts to Prove the Conjecture

The statement of Comon's Conjecture can now be stated in more formal notation, introduced in the previous chapter.

Conjecture (Comon, (Comon et al.[13])). *Given a symmetric, order d tensor $T \in S^d(\mathbb{F}^n) \subset (\mathbb{F}^n)^{\otimes d}$, then **Rk**$(T) =$ **SRk**(T).*

The ability to rewrite symmetric tensors of fixed order both as homogenous polynomials (analogous to symmetric tensor decomposition), and as a sum of regular tensors was the key factor that seems to have inspired this conjecture. This section details some of the papers that prove that the conjecture holds under certain constraints.

As mentioned in the introduction Comon et al. [13] showed that the conjecture was true for tensors of "large enough" order — a notion that became concrete in later literature. The authors noted that, as a result of the Veronese embedding (a canonical construction in algebraic geometry), the rank of a symmetric tensor $T \in S^d(\mathbb{C}^n)$ cannot be more than $\binom{n+d-1}{d}$. Also, since $S^d(\mathbb{C}^n) \subseteq (\mathbb{C}^n)^{\otimes d}$, they pointed out that the following was true for all $T \in S^d(\mathbb{C}^n)$: that the **Rk**$(T) \leq$ **SRk**(T). Their major result was that the conjecture holds "generically" (i.e in a dense subset) whenever **SRk**$(T) \leq n$, the dimension of the underlying vector space. In particular, for every $T \in S^d(\mathbb{C}^n)$, they showed that the conjecture held if the **SRk**$(T) = 1$ or 2.

A concurrent approach to tackling Comon's conjecture arises from the transposition of classical problem from number theory — the Waring decomposition of a number — to the language of polynomials. This famous problem then gets restated as follows:

What is the the minimum number of linear polynomials required such that a linear combination of them to the dth power is equal to a homogenous degree-d polynomial? In mathematical notation, with $f \in \mathbb{F}[x_1, \ldots, x_n]_d$, a homogenous polynomial of degree d and $\mathbf{a} := (a_1, \ldots, a_n)$ denoting a vector containing coefficients such that $\mathbf{a} \cdot \mathbf{x} = a_1x_1 + \cdots + a_nx_n$.

$$\min_{s>0} \sum_{|\alpha|=d} f_\alpha \mathbf{x}^\alpha = \sum_{i=1}^{s} (\mathbf{a}_i \cdot \mathbf{x})^d \tag{3.1}$$

A solution to the polynomial Waring problem solves the issue of computing symmetric rank for a given symmetric tensor (see Section 2.3). In fact this relationship is so significant that in the literature, the term **Waring Rank** is used synonymously with symmetric rank when discussing symmetric tensors. An equivalent formulation for this problem is in terms of predicting the dimensions of higher secant varieties and relating these predictions to the Waring decompositions for symmetric tensors [4]. The polynomial Waring problem was solved by Alexander and Hirschowitz in 1995 by using this secant variety formulation, building on previous work from Campbell, Palatini and Terracini, and using the novel "differential Horace's method" [4]. They showed that $\sigma_r(\nu_d(\mathbb{PF}^{n+1}))$ - the r-th secant variety to the d-th Veronese embedding of $\mathbb{F}^{n+1}$ - is of expected dimension with 5 exceptional cases. Brambilla and Ottaviani [6] in 2008 gave a self-contained proof of the theorem, with some simplifications, particularly in the case of order 3 symmetric tensors. The existence of concrete methods for computing rank and symmetric rank meant that for any given symmetric tensor, Comon's conjecture could be quickly verified or disproved.

Ballico and Bernardi [2] in 2013 showed that that a tensor T lying on the tangent variety to a Veronese variety must always have $\textbf{Rk}(T) = \textbf{SRk}(T)$. Their paper highlighted explicit algorithms for the computation of the rank of a tensor that could be expressed as the limit of a linear combination of elementary tensors. The paper showed an application to Comon's conjecture occurred when working specifically with symmetric tensors.

While secant varieties are convenient geometric tools to use for computing rank and symmetric rank via the AH theorem, Buczynska and Buczynski in 2013 [7] demonstrated how the commonly used practice of using the $(r+1) \times (r+1)$ minors of the *catalecticant matrix* to compute the generating equations for the secant variety are not sufficient for higher order Veronese varieties (order $>> 2$). This paper resulted in the developement of *cactus varieties*.

Computing decompositions covers the first question about symmetric rank — the existence of a minimal length symmetric decomposition. The other question, concerning the uniqueness of a decomposition, requires the definition of when a tensor is *identifiabililty*.

Definition 3.1.1 (Identifiability)**.** A tensor $T \in V_1 \otimes \cdots \otimes V_d$ is said to be **identifiable** if it admits a unique rank decomposition (modulo scaling by d-th roots of unity and permutations of summands)

Example 4. The tensor $(3,4,0) \otimes (1,1,0) + (4,5,0) \otimes (1,0,0)$ can be written as $(7,9,0) \otimes (1,0,0) + (3,4,0) \otimes (0,1,0)$, and is thus not identifiable, since it can be decomposed in distinct ways, not involving simply a permutation of summands or a direct rescaling.

Identifiability is a useful property when working with tensors because no further computation is required once a rank decomposition has been found for a given identifiable tensor.

Generic identifiability is the idea that almost every tensor (i.e. a tensor in a dense open subset), is identifiable. Specific identifiability poses the opposite question - given a specific tensor T, does it admit a unique decomposition? Chiantini, Ottaviani and Vannieuwenhoven [11, 12] in two subsequent papers tackled both of these questions. Their earlier paper showed that a general symmetric tensor in $S^d(\mathbb{C}^{n+1})$ of sub-generic rank r was identifiable, provided $r \leq \binom{n+d}{d}/(n+1)$ (with all exceptions again being covered by the Alexander-Hirschowitz theorem). Their second paper focused on showing what criteria could be called "effective" to determine if a given tensor was specifically identifiable. Criteria were designated effective if their conditions were satisfied by generic points (i.e. the criteria were satisfied in an open, dense subset). One of the conclusions of this paper was that Kruskal's criterion fell in line with this definition of *effectiveness*.

Friedland [15] in 2016 focused not on the computation of the symmetric rank, but on the properties of symmetric tensors of a given symmetric rank. In particular the paper focused on showing that when a symmetric tensor T can be flattened into a matrix $T^{(i)}$ (see Section 4.1.4), then Comon's Conjecture held if the rank of the tensor, **Rk**$(T) =$ **Rk**$(T^{(i)})$ or if **Rk**$(T) =$ **Rk**$(T^{(i)})$. Zhang et al. [28] conclusively showed that Comon's conjecture held for all tensors where the rank of the tensor under consideration is not more than its order.

Ballico et al [3] at the end of 2018, refocused their efforts on the question of identifiability, and worked on refining Kruskals criterion (sharp and algebraic in nature) into a geometric criterion that can be verified more easily. They concluded that Comon's conjecture held for tensors whose symmetric rank r is strictly bounded above by $\binom{n+e}{e}$ for a specific choice of e. This choice depended upon splitting the Segre product (another classical construction from algebraic geometry) into two disjoint components and then imposing conditions on a set of points on the Segre variety that exhibited distinct behaviour when projected down to the first component versus the second. The authors of this paper clearly pointed out that even though they had increased the scope of validity of Comon's conjecture, Shitov's proposed counterexample [22] lay outside this scope and thus might have still been a plausible counterexample.

Bernardi et al. [4] in 2018 comprehensively covered the literature and algorithms required for the application of secant varieties to questions about (generic) tensor decomposition, with a detailed explanation of the case of symmetric tensor decomposition and its relationship to the AH theorem.

Casarotti et al. [10] in 2018 proposed to improve the bounds for Comon's conjecture when working with flattenings, provided that equations for the secant varieties associated to the Veronese variety could be generated. Their result was for a tensor T having rank $r < \binom{n+\lfloor \frac{d}{2} \rfloor}{n}$, the **SRk**$(T) = r$, for cases where such generating determinantal equations were known.

Finally in a paper from 2020, Seigal [21] demonstrated that for all "cubic surfaces", Comon's conjecture does hold. Here cubic surfaces were used to refer to the fact that under the identification with homogenous polynomials, order 3 symmetric tensors correspond to cubic polynomials, and thus their zero sets could be called cubic surfaces. The author also used flattenings of tensors (see Section 4.1.4), dividing their proof into cases depending on the number of singularities on the cubic surfaces in question.

3.2 Overview of Valid Cases of the Conjecture

The current overview of cases where Comon's Conjecture is known to hold is as follows:

- For $T = \sum_{i=1}^{s} y_i^{\otimes d}$ and d *large enough* [13]. The large enough argument in the paper hinges on a classical result from algebraic geometry, that d points in $\mathbb{C}^n$ form the solution set of of polynomial equations of degree $\leq d$
- For $T \in S^d(\mathbb{C}^n)$ and **SRk**$(T) = 1, 2$ [13]
- For $T \in S^d\mathbb{F}^n$, and **Rk**$(T) \leq \frac{dn-d+1}{2}$ where $\mathbb{F} = \mathbb{R}, \mathbb{C}$ [19]
- For $T \in S^d(\mathbb{C}^n)$ for $(d, n) \in \{(3, 2), (4, 2), (3, 3)\}$ [15]
- In the case $d \geq 3$, char$(\mathbb{F}) \geq 3, T \in S^d(\mathbb{F}^n)$ and **Rk**$(T) =$ **Rk**$(T^{(i)})$ or **Rk**$(T^{(i)}) + 1$ (Here $T^{(i)}$ denotes the flattening (see Section 4.1.4) of T along any mode. Since T is symmetric all directions are equivalent) [15]
- When $T \in S^3(\mathbb{C}^n)$ and **Rk**$(T) \leq 5$ or **SRk**$(T) \leq 6$ [15]
- For $T \in S^d(\mathbb{F}^n)$ where **Rk**$(T) \leq d$ where $\mathbb{F} = \mathbb{R}, \mathbb{C}$[28]
- If T is a tensor lying on a tangential variety to a Veronese variety [2]
- If $T \in \mathbb{C}^2 \otimes \mathbb{C}^n \otimes \mathbb{C}^n$ [8]
- For T being a Coppersmith-Winograd tensor [17]
- Whenever a tensor T is cubic surface (real or complex) i.e. T can be represented by a homogenous cubic polynomial $\sum_{|\alpha|=3} \lambda_\alpha \mathbf{x}^\alpha$ [21]

Chapter 4

Constructions required for a Counterexample

4.1 Background to the Counterexample

In spite of all of these preliminary results indicating positive answers towards Comon's Conjecture, Yaroslav Shitov seemed to arrive at a method of constructing a potential counterexample, as early as 2018 [22]. An error found in this paper led to the invalidation of this specific counterexample in 2024, but the period of six years in the interim led to multiple other preprints from Shitov [23–25].

These preprints started with a newer, alternate methodology for the construction of counterexamples. While there is now no material carrying the stamp of peer-review and Shitov's name, his initial contribution and subsequent preprints seem to indicate a viable approach to constructing a counterexample to Comon's conjecture over $\mathbb{R}$, if not $\mathbb{C}$.

4.1.1 Shitov's First Paper in 2018

The 2018 paper began with taking the ground field as $\mathbb{C}$, and fixing the notation to be used. The main objects of interest in the paper were order-3 tensors, and the introductory section of the paper highlighted the behaviour of such tensors under elementary transformations with respect to their constituent order-2 "slices" (see Section 4.1.3). The construction of the counterexample began with the description of a 100×100 matrix, partitioned into 20 submatrices. Each submatrix was then mapped to a corresponding 100×100 matrix, now with entries having the same indices as the submatrix set to 1 and all other entries as 0. Taking the $1-$ and $2-$ supports of each of these, and pre- and post-multiplying them with a column-shift operator with parameter a led to another set of matrices of the same size (still 100×100).

It is at this point that the paper made a concerning construction. A set $\mathcal{B}$ was constructed consisting of matrices of distinct formats (viz. 200×200 and 400×400). While there is no issue in defining such a set, the paper then constructed "the $\mathbb{C}$-linear span" of the elements of this set. Addition of quantities that do not live in the same dimensional vector space, without a specified or implied embedding map (at least from this author's prior knowledge) presented a major problem with following this paper.

Later in the paper a tensor decomposition was described (Eq. 5.6 in [22].) This statement was the subject of the erratum published in 2024 [14]. The erratum indicated such a statement was true only in specific cases, but could not be said to hold in general otherwise without proof. Proving the statement has been a non-trivial challenge, and since no answer has been arrived at, the current consensus is that the paper [22] does not lead to a valid counterexample.

While certain choices were made during this process that led to unanswered questions, the overall tools for later establishing a valid counterexample were first established in this paper. These included the two minor operations - slicing tensors (Section 4.1.3) and flattening tensors (Section 4.1.4) and the three major operations - adjoining slices to tensors (Section 4.2.1), generating families modulo slices (Section 4.2.3) and cloning (Section 4.2.4).

4.1.2 Shitov's Subsequent Preprints

In the following years, the preprints [23–25] from Shitov either tackled Comon's conjecture in a restricted setting, or developed tools to construct more general counterexamples. These might also be used to construct a counterexample to Strassen's conjecture (another tensor rank conjecture concerning the additivity of rank). The preprint from 2020 [23] tackled the conjecture over $\mathbb{R}$ and it was this preprint that formed the basis for the paper from Wang and Seigal [27].

The constructions made in each preprint followed the same pattern, indicating a common toolset and structure of approach. Each approach saw an order-d tensor as composed of lower order-$(d-1)$ tensors, which allowed for tensors to be transformed per "slice" along a given mode. Viewing tensors in this manner opened a way to describe tensors modulo these slices — a useful operation, seen in both number theory and algebraic geometry as a tool to study problems in a more localised setting. Such a family, when generated, gave an idea of the general properties of the orginal tensor (how many linearly independent slices along a specified mode, maximum slice rank along each mode etc.) Flattenings assisted with the computation of rank for these slices (because computing the rank of matrices is a known and easier problem to solve than doing so for higher order tensors) and the flattenings lent themselves well to describing rank information that might change in the relationship between a tensor and its slices.

These tools allowed for an analysis of any particular tensor, but the "construction" aspect of this approach became evident with the adjoining of slices to a tensor and the cloning operation. The former operation changed the format of the original tensor with respect to certain "chosen" slices, and allowed the rank information of the family generated modulo the slices to be preserved in a new tensor containing both the original and the slices that were required.

Cloning was a way to maintain rank information by simply making a higher format copy, and allowed for more "freedom" when modifying slices — simply because there were more to chose from. Cloning started to play a role because perturbing such an adjoined tensor (usually by the modification of one of the constituent slices) caused a violation of the properties that were supposed to be preserved.

While each of these preprints started with this general structure, and [25] proposed the construction of a general counterexample over any $\mathbb{F}$ with char$(\mathbb{F}) \neq 3$, an intermediate step involved in all of these was the construction of an "emulating family". Such a family, if it existed, would allow for the rest of the constructions to proceed. However, proving the existence of such a family was still not trivial. The latest preprint attempted to do so, highlighting a similar construction from [24], but tracing the construction back led to the definition of a function dependent on an arbitrary constant - in the same way as the paper [22] relied on the existence of an arbitrarily partitioned 100×100 matrix. Such choices, while perhaps valid, did not appear to be explained at any point in their corresponding documents, raising some skepticism and prompting further investigation into better documented methods that could explain, or proofs that could deny it .

4.1.3 Slicing a Tensor

Viewing the columns or rows of a matrix as linearly dependent or independent and how they affect the overall linear transformation described by the matrix is the exact low-order analogue for describing tensors as composed of lower-order slices. Conventionally the columns of a matrix would be the 1-slices, and the rows 2-slices

Definition 4.1.1 (Slices of a Tensor)**.** Let $T := v_1 \otimes \cdots \otimes v_d \in V_1 \otimes \cdots \otimes V_d$ be an elementary, order d tensor. Here each $v_i \in V_i$, with V_i a vector space with dimension n_i.

Then, **the j-th i-slice of T** is given by

$$T^i_j := (v_i)(j)(v_1 \otimes \cdots v_{i-1} \otimes v_{i+1} \cdots \otimes v_d) \in V_{\hat{i}}.$$

Here $1 \leq j \leq n_i$. For each *mode* $1 \leq i \leq d$, there are n_i order-$d-1$ slices that make up T.
This definition is equivalent to selecting the j-th i-slice of the d-way array that represents T. For a general tensor $T \in V_1 \otimes \cdots \otimes V_d$, the slice T^i_j is exactly the sum of the jth i-slices of elementary tensors that make up a rank decomposition of T.

Observation. Through the rest of this thesis, the superscript accompanying a tensor variable indicates the chosen mode along which a specified operation must occur.

Example 5. Consider a tensor $T \in \mathbb{F}^4 \otimes \mathbb{F}^3 \otimes \mathbb{F}^2$, where $T := (1,2,3,4) \otimes (1,2,3) \otimes (1,2)$. Then slices along different modes are computed as follows (visualised in Figure 3)

- $T^1_3 = 3((1,2,3) \otimes (1,2))$
- $T^2_2 = 2((1,2,3,4) \otimes (1,2))$
- $T^3_2 = 2((1,2,3,4) \otimes (1,2,3))$

Observation. The operation defined above for taking the j-th i-slice of an order d tensor, $S^i_j : V := V_1 \otimes \cdots \otimes V_d \to V_{\hat{i}}$ defined as $T \mapsto T^i_j$ is a well-defined linear mapping between vector spaces, by Definition 4.1.1.

Lemma 4.1.1. *Given a tensor T such that* ***Rk***$(T) = r$*, then there is at least one mode $1 \leq i \leq d$ having r linearly independent slices T^i_j where $1 \leq j \leq n_i$.*

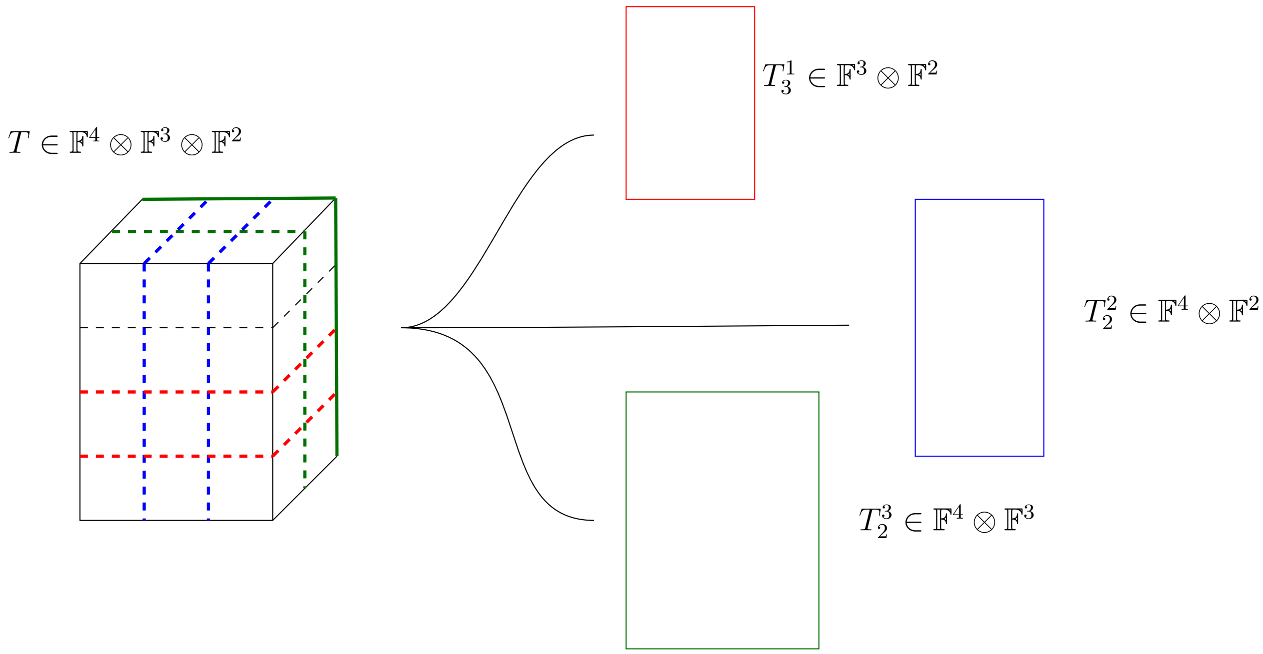

Figure 3: Slicing a tensor T along different modes

Proof. Suppose the claim is false i.e. there is no mode i such that the number of linearly independent slices T_j^i is r. Since we are only dealing with a finite order d, there is a mode i with the largest number of linearly independent slices T_j^i. Let this number be ρ.

$\rho < r$: Then each of the slices $S_i^j(T)$ can be written as a linear combination of exactly ρ slices of order $(d-1)$. One can therefore rewrite the decomposition of T in terms of vectors in V_i and the linearly independent set of $S_j^i(T)$, a decomposition of length less than r, contradicting the definition of rank.

$\rho > r$: Then any decomposition of length r must miss out at least one of the linearly independent slices that make up T, a contradiction to the definition of a tensor decomposition.

Thus the assumption of the claim being false leads to a contradiction. □

Corollary 4.1.1.1. *Given a tensor T of rank r, the collections of slices $\{T_j^i\}_{j=1}^{n_i}$ along all modes $1 \leq i \leq d$ have r linearly independent elements, i.e $\langle T_j^i \rangle_{j=1}^{n_i}$ is an r-dimensional subspace.*

A major claim from the literature on Comon's conjecture is the following stronger theorem about the number of linearly independent slices along a chosen mode. The theorem as stated below first appeared in [28], but in a more geometric formulation it was proved in [8].

Theorem 4.1.2 ([28], [8])**.** *Let T be an order d tensor of rank r. Then for any mode i, the set of slices $\{T_j^i \in V_{\hat{i}} | 1 \leq j \leq n_i\}$ is linearly independent.*

When working with symmetric tensors, slicing along any mode results in a lower order tensor in the same "format" (referring to the dimension of the underlying vector space). In this case $S_j^i : (V)^{\otimes^d} \to (V)^{\otimes^{d-1}}$ for all modes i and all slices $1 \leq j \leq n_i$.

4.1.4 Flattening a Tensor

Flattening, while not as important in Shitov's constructions, is an useful operation to verify rank information for tensors. Such an operation can also be used as a quick check when performing operations or constructions on tensors to detect changes in rank. This is due to the fact that computing the rank of a matrix is a much easier problem than computing the rank of an order d tensor, especially as d increases.

Definition 4.1.2 (Flattening a Tensor)**.** Let $T \in V_1 \otimes \cdots \otimes V_d$, be an order d tensor. Define $T^{(i)} \in \left(\bigoplus_{j \neq i} V_j\right) \otimes V_i$, the i-th flattening as follows

$$T^{(i)}((k_1, \ldots k_{i-1}, k_{i+1} \ldots k_d), k_i) := T(k_1, k_2, \ldots, k_d)$$

As with slicing, this definition is equivalent to flattening the d-way array representing T into a matrix with the elements in a row indexed by $[n_i]$ and elements in a column indexed by $\times_{j \neq i}[n_j]$, in lexicographic order.

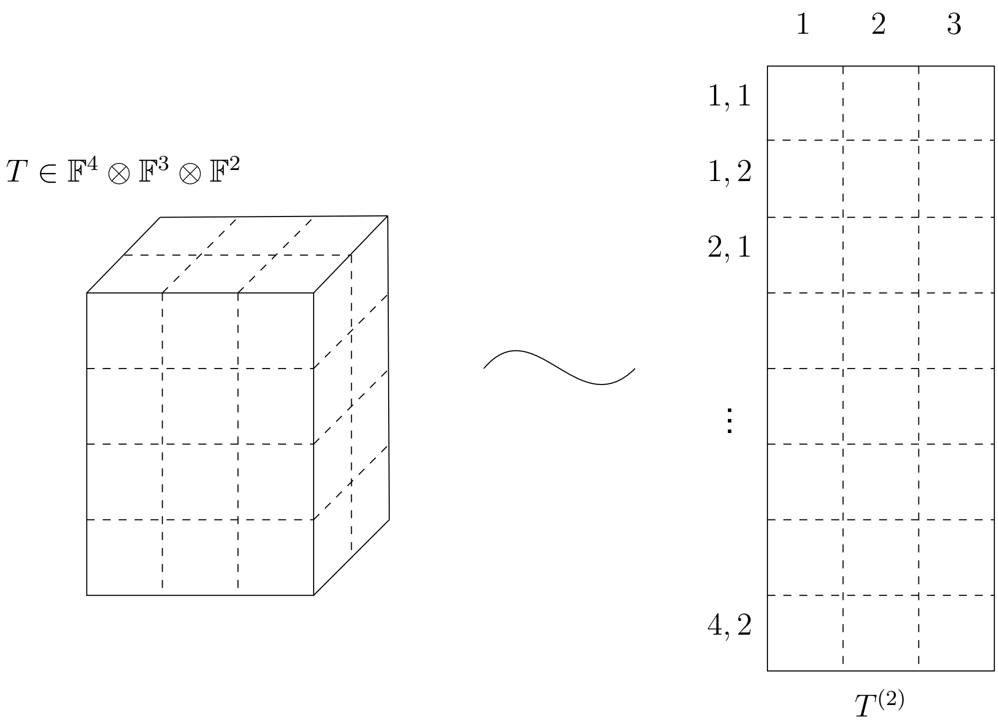

Figure 4: Flattening a tensor T

Example 6. Consider a tensor $T \in \mathbb{F}^4 \otimes \mathbb{F}^3 \otimes \mathbb{F}^2$, where $T := (1,2,3,4) \otimes (1,2,3) \otimes (1,2)$. Then flattening the tensor along the 2-mode can be computed as

$$T^{(2)} = ((1,2,3,4) \otimes (1,2))^{(1)} \otimes (1,2,3).$$

While this looks strange, the vector required to be tensored with $(1,2,3)$ is formed by flattening the matrix $(1,2,3,4) \otimes (1,2)$ along the 1-mode. This is equivalent to the vector $(1,2,3,4,2,4,6,8)$. Figure 4 provides an illustration of this example, also highlighting how the column and row indices are generated.

Observation. Flattening T to $T^{(i)}$ is the operation by which T could be seen as a linear map between vector spaces instead (a modification of the illustration in Section 2.1). Here $T^{(i)} \in V_{\hat{i}} \otimes V_i$, seen as a matrix, representing a mapping from $V_i \to V_{\hat{i}}$ Each column $T_j^{(i)} := T^{(i)}(\dots|j)$ represents the *unfolding* of the j-th i slice of T, denoted $(T_j^i)^{\emptyset}$, into a vector.

Observation. The flattening operation $F^{(i)} : V := V_1 \otimes \cdots \otimes V_d \to W \otimes V_i, T \mapsto T^{(i)}$ is a well defined linear mapping between vector spaces where $W = \bigoplus_{j \neq i} V_j$.

Remark. In the literature surveyed, flattening can be done not just along one mode, but for a collection of modes. This gives rise to the empty set notation denoting the unravelling of a vector. Since this is less important for the following constructions, this more general definition of flattening is not covered. However, in this document, the spirit of such a general operation is retained by also describing the unfolding of an order d tensor, seen in the following remark.

Remark. For a tensor $T \in V_1 \otimes \cdots \otimes V_d$, the unfolding $T^{\emptyset}$ denotes a vector lying in $V_1 \oplus \cdots \oplus V_d$. Here each component of the vector is indexed by the the set $\times_{i=1}^{d} [n_i]$, and $T^{\emptyset}((k_1, \dots, k_d)) = T(k_1, \dots, k_d)$.

When dealing with symmetric tensors, flattening along any mode is equivalent (since the tensor is symmetric), and so $T^{(i)} = T^{(j)}$ for all $i \neq j$. The resulting matrix is an element of $\mathbb{F}^{(d-1)n} \otimes \mathbb{F}^n$.

4.2 Major New Constructions

While the preceding constructions are relatively ”minor” in the grand scheme of Shitov's proposed counterexample to Comon, the following three major operations on tensors are integral to the final construction proposed (and indeed implemented by Wang and Seigal [27]). The following subsections deal with constructions involving the operations on a specific tensor parametrised by families of tensors of lower order and appropriate format. It is convenient to represent such families of slices in terms of an appropriate order-d tensor, and then reframe definitions accordingly in terms of this newly created order-d tensor.

Definition 4.2.1 (Creating a Tensor from a Family of Slices)**.** Let $d > 0$ and $1 \leq i \leq d$. Take $\mathcal{M} := \{M_1, \dots, M_N\} \subset V_{\hat{i}}$ as a finite collection of order-$(d-1)$ tensors. Since $\mathcal{M}$ is finite, it can be rewritten as a tensor of order d, $\bar{\mathcal{M}} \in V_1 \otimes \cdots \otimes V_{i-1} \otimes \mathbb{F}^N \otimes V_{i+1} \otimes \cdots \otimes V_d$, defined in terms of each of its i-slices $\bar{\mathcal{M}}_j^i$, where $1 \leq j \leq N$,

$$\bar{\mathcal{M}}(k_1, \dots k_{i-1}, j, k_{i+1} \dots, k_d) := M_j(k_1, \dots k_{i-1}, k_{i+1} \dots k_d).$$

4.2.1 Background to Adjoining

Once the notion of a tensor of order d being composed of various order $(d-1)$ slices has been established, one natural operation that can be performed is the *adjoininig* of slices along the d modes of the tensor, provided that each slice is of appropriate format. To perform this operation, one needs two pieces of information — the tensor to be transformed, and the collections of slices (of appropriate format) that are to be adjoined. This operation seems to have been first formally defined by Shitov in his 2018 paper [22], and further developed into the language below in his preprint from 2020[23].

In this paper, the operation involving only one mode is discussed, before reframing the operation involving all the modes as the result of a series of successive operations. Doing so requires the tensor to be first embedded into the tensor product of $(d-1)$ finite dimensional vector spaces and the following direct sum of the field $\mathbb{F}$ (denoted $\bigoplus_{\lambda\in\Lambda}\mathbb{F}$, where Λ is an indexing set). To work with this embedding that captures all the information required, it is then necessary to project back down to a tensor product of finite-dimensional vector spaces. Finally when moving between modes, since the lower-order tensors chosen were appropriately sized for the original tensor, a padding function is required to continue the adjoining process, without adding any extraneous information.

Definition 4.2.2 (Adjoining Slices to a Tensor along a specific Mode)**.** Let $T \in V_1 \otimes \cdots \otimes V_d$ be an order d tensor, and $\mathcal{M} = \{M_1, \dots, M_N\} \subset V_{\hat{i}}$ be a finite collection of order-$(d-1)$ slices.

Naively, first define the required image of the adjoining map $\tau := Adj^i(T, \bar{\mathcal{M}}) \in V_1 \otimes \cdots \otimes V_{i-1} \otimes \left(V_i \oplus \mathbb{F}^N\right) \otimes V_{i+1} \otimes \cdots \otimes V_d$ in the following manner

$$\tau(k_1, \dots, k_d) = \begin{cases} T(k_1, \dots, k_d) & (k_i)_{i=1}^d \in \times_{i=1}^d [n_i] \\ \bar{\mathcal{M}}(k_1, \dots, k_d) & (k_j)_{j\neq i} \in \times_{j\neq i}[n_j] \text{ and } n_i \leq k_i \leq n_i + N \end{cases}$$

Note that the above description is still coordinate dependent. Define $\hat{V}_i := V_1 \otimes \cdots \otimes V_{i-1} \otimes \left(\bigoplus_{\lambda\in\Lambda}\mathbb{F}\right) \otimes V_{i+1} \otimes \cdots \otimes V_d$. Defining $\hat{V}_i$ in this manner allows for the following (re)definition of the adjoining operation, now in terms of a well-defined function:

$$Adj^i : V_1 \otimes \cdots \otimes V_d \times \hat{V}_i \to \hat{V}_i,$$

taking $(T, M) \mapsto \tau$ as described above. The only modification to the description of τ is now that $\tau(k_1, \dots, k_d) = 0$ for any values of $k_i > N + n_i$.

It becomes necessary to define an appropriate projection mapping, to allow the image of this map to be viewed as a tensor in finite dimensions along every mode.

Definition 4.2.3 (Projecting to obtain a tensor of finite format)**.** Given an element $T \in \hat{V}_i$, define the projection mapping

$$\pi_N^i : \hat{V}_i \to V_1 \otimes \cdots \otimes V_{i-1} \otimes \mathbb{F}^N \otimes V_{i+1} \otimes \cdots \otimes V_d$$

as restricting T' to an order d tensor of finite format in a natural way. In coordinates

$$\pi_N^i(T)(k_1, \dots, k_d) = T(k_1, \dots, k_d), (k_i)_{i=1}^d \in [n_1] \times \cdots \times [n_{i-1}] \times [N] \times [n_{i+1}] \times \cdots [n_d]$$

Remark. The above two definitions allow for obtaining a tensor of finite format when adjoining a family of slices along only one mode. The next major step to be taken is adjoining slices along multiple directions. Doing so requires the definition of a ”padding” operation, that allows for order-$(d-1)$ tensors to be adjusted into appropriate format, before being adjoined in the appropriate direction.

Definition 4.2.4 (Padding an order-$(d-1)$ tensor without increasing order)**.** Given $d > 0$ $1 \leq i \leq d$ as fixed. Let $T \in V_1 \otimes \cdots \otimes V_d$ be an order d tensor, and $A \in W_1 \otimes \cdots \otimes W_{i-1} \otimes W_{i+i} \otimes \cdots \otimes W_d =: W_{\hat{i}}$ be an order $d-1$ tensor, such that $\dim(V_j) \geq \dim(W_j)$ for all $j \neq i$. Then

$$pad^i : W_{\hat{i}} \times V \to V_{\hat{i}}$$

as a function can be defined in coordinates as

$$pad^i(A,T)(k_1 \dots k_{i-1}, k_{i+1} \dots k_d) = \begin{cases} A(k_1 \dots k_{i-1}, k_{i+1} \dots k_d) & 1 \leq k_j \leq \dim(W_j) \forall j \neq i \\ 0 & \text{otherwise} \end{cases}$$

Combining all of these functions (Adj^i, π^i_N and pad^i) allows for adjoining families of $d-1$-order tensors along every mode of a given order d tensor according to the following procedure.

4.2.2 Adjoining Slices to a Tensor

Given an order d-tensor $T \in V$ and a collection of finite subsets of $d-1$ tensors $\mathcal{M}_i \subset V_{\hat{i}}$ for $1 \leq i \leq d$. Then $Adj(T, \mathcal{M}_1, \dots \mathcal{M}_d)$ can be constructed in the following manner

Step 1 $T_1 = \pi^1_{|\mathcal{M}_1|+n_1} \circ Adj^1(T, \bar{\mathcal{M}}_1)$ resulting in a new intermediate tensor with slices adjoined along the 1-mode, of finite format.

Step 2a Let $M_2 = \{pad^2(A, T_1) | A \in \mathcal{M}_2\}$, creating a set of appropriately padded slices, now to be adjoined along the 2-mode.

Step 2b $T_2 = \pi^2_{|M_2|+n_2} \circ Adj^2(T_1, \bar{M}_2)$

⋮

Step da $M_d = \{pad^d(A, T_{d-1}) | A \in \mathcal{M}_d\}$

Step db $T_d = \pi^d_{|M_d|+n_d} \circ Adj^d(T_{d-1}, \bar{M}_d)$

The tensor $T_d \in (V_1 \oplus \mathbb{F}^{|\mathcal{M}_1|}) \otimes \cdots \otimes (V_d \oplus \mathbb{F}^{|\mathcal{M}_d|})$ is the required result of the process, $Adj(T, \mathcal{M}_1, \dots, \mathcal{M}_d)$.

Observation. Figure 5 illustrates the above definition for the case when only one slice is being adjoined in one direction. According to the above definition τ can be seen as an order d tensor still, but now lying in $(V_1 \oplus \langle e_{1,n_1+1} \rangle) \otimes \cdots \otimes V_d$ where e_{1,n_1+1} indicates a ”new” basis vector added onto the original vector space V_1.

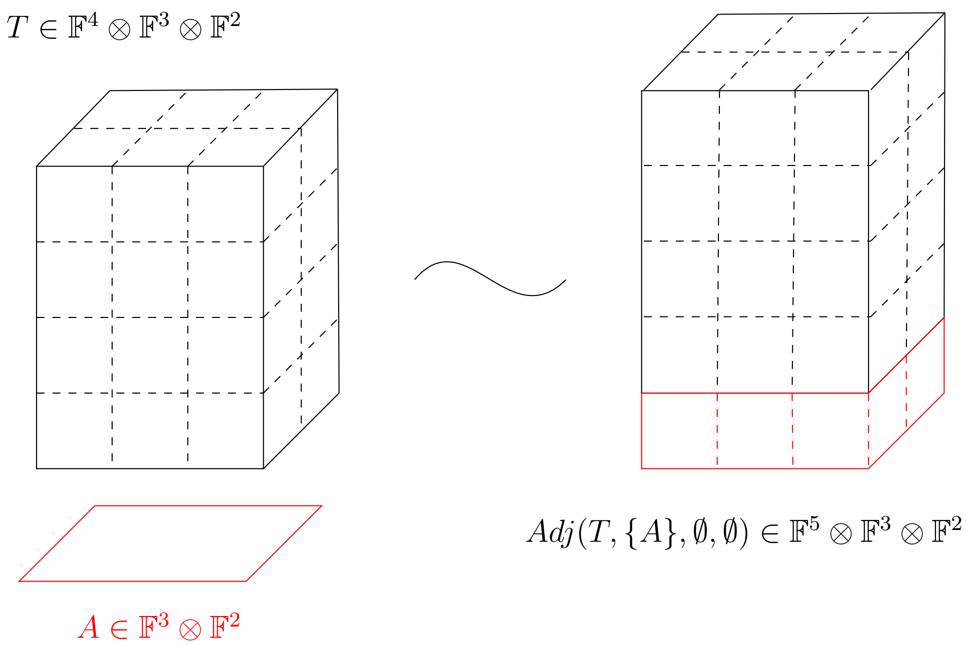

Figure 5: Adjoining a slice A to a tensor T

The *symmetric adjoin* can be denoted $SAdj(T, \mathcal{M})$ where $\mathcal{M}$ is adjoined along every mode, following the same process as given in Section 4.2.2.

The adjoining process as described creates a tensor of larger format, but the rank of such a tensor may be appropriately constrained depending on the slices being adjoined. It is this constraining property that Shitov [25] and Wang and Seigal [27] use to construct an intermediate tensor with adjoined slices that has a differing rank and symmetric rank.

Remark. It may be trivial to note that while the order of the slices in $\mathcal{M}_i$ matters when constructing the final tensor, the rank information of the constructed tensor is independent of such changes in ordering. Further, the adjoining procedure itself can be modified, to follow any ordering of modes, as long as the slices required have been padded appropriately.

4.2.3 Generating Tensor Families modulo Slices

Given that a tensor can admit many non-unique decompositions, one way to analyse the structure of a tensor is via an analogy to Gaussian elimination for matrices. The substitution method as described in [1, 17] has been a computational tool for analysing tensors since the 1960's. Rather than arriving at one possible "lowest" optimal tensor, generating a family of tensors modulo a set of slices is equivalent to looking at all the possible ways a tensor could be transformed, based on transforming its constituent slices along different modes.

Defining such an operation that generates a linear span necessitates one additional, alternate description of padding a tensor into an appropriate format, now in terms of seeing the lower order slice as embedded into an empty higher order tensor. In doing so, it is also necessary to take a candidate higher order tensor as a "template" that the slice can be made to resemble.

Definition 4.2.5 (Padding an order-$(d-1)$ tensor into higher order)**.** Let $d > 0$, and $1 \leq i \leq d$. Let $T \in V_1 \otimes \cdots \otimes V_d$ be the candidate higher order tensor to be used as a template, and $A \in V_{\hat{i}}$ the order $d-1$ tensor to be padded into order d.

Then define $Pad^i_j : V_{\hat{i}} \times V \to V$ in coordinates as follows

$$Pad^i_j(A,T)(k_1,\dots,k_d) = \begin{cases} A(k_1,\dots,k_{i-1},k_{i+1},\dots k_d) & k_i = j \\ 0 & \text{otherwise} \end{cases}$$

Equipped with this method of padding one can now define slicing a tensor T along one particular mode.

Definition 4.2.6 (Tensor Family Generated by Slices along a Mode)**.** Let $d > 0$ and $1 \leq i \leq d$. Let $T \in V := V_1 \otimes \cdots \otimes V_d$ be an order d tensor, and $\mathcal{M} := \{M_1,\dots,M_N\} \subset V_{\hat{i}}$ be a finite collection of order-$(d-1)$ slices of the appropriate format.

Define $Mod^i(T,\bar{\mathcal{M}}) : V \times V_1 \otimes \cdots \otimes V_{i-1} \otimes \mathbb{F}^N \otimes V_{i+1} \otimes \cdots \otimes V_d \to 2^V$ in terms of coordinates as follows (2^P denoting the power set of a set P):

$$Mod^i(T,\bar{\mathcal{M}}) = \left\{ T + \sum_{k=1}^{n_i} \sum_{j=1}^{N} \lambda_{j,k} Pad^i_k(S^i_j(\bar{\mathcal{M}})) \mid \lambda_{j,k} \in \mathbb{F} \right\}.$$

Here slices are first selected from the constructed order d tensor, then padded appropriately to remain of the same format as T, then scaled and added to T.

This definition can now be extended to performing this operation along all the d modes. The only difference now notationally is writing $Mod^j(Mod^i(T,\bar{\mathcal{M}}_i),\bar{\mathcal{M}}_j)$ to denote the $Mod^j(S,\bar{\mathcal{M}}_j)$ for every $S \in Mod^i(T,\mathcal{M}_i)$ and then taking the union of all the resulting sets. Thus the final family $F := T \operatorname{mod}(\mathcal{M}_1,\dots \mathcal{M}_d)$ is a result of the d-fold composition of the distinct Mod^i operations.

Proposition 4.2.1. $Mod^j(Mod^i(T,\bar{\mathcal{M}}_i),\bar{\mathcal{M}}_j) = Mod^i(Mod^j(T,\bar{\mathcal{M}}_j),\bar{\mathcal{M}}_i)$

Proof.

$$\begin{aligned}
& Mod^j(Mod^i(T,\bar{\mathcal{M}}_i),\bar{\mathcal{M}}_j) \\
= \; & \left\{ Mod^j(S,\bar{\mathcal{M}}_j) | S \in Mod^i(T,\bar{\mathcal{M}}_i) \right\} \\
= \; & \left\{ S + \sum_{a=1}^{n_j} \sum_{b=1}^{N_j} \lambda_{a,b} Pad^j_a(S^j_b(\bar{\mathcal{M}}_j)) | \lambda_{a,b} \in \mathbb{F}, S \in Mod^i(T,\bar{\mathcal{M}}_i) \right\} \\
= \; & \left\{ T + \sum_{c=1}^{n_i} \sum_{d=1}^{N_i} \lambda_{c,d} Pad^i_c(S^i_d(\bar{\mathcal{M}}_i)) + \sum_{b=1}^{n_j} \sum_{a=1}^{N_j} \lambda_{a,b} Pad^j_a(S^j_b(\bar{\mathcal{M}}_j)) | \lambda_{a,b}, \lambda_{c,d} \in \mathbb{F} \right\} \\
= \; & \left\{ S + \sum_{c=1}^{n_i} \sum_{d=1}^{N_i} \lambda_{c,d} Pad^i_c(S^i_d(\bar{\mathcal{M}}_i)) | \lambda_{c,d} \in \mathbb{F}, S \in Mod^j(T,\bar{\mathcal{M}}_j) \right\} \\
= \; & \left\{ Mod^i(S,\bar{\mathcal{M}}_i) | S \in Mod^j(T,\bar{\mathcal{M}}_j) \right\} \\
= \; & Mod^i(Mod^j(T,\bar{\mathcal{M}}_j),\bar{\mathcal{M}}_i)
\end{aligned}$$

$\square$

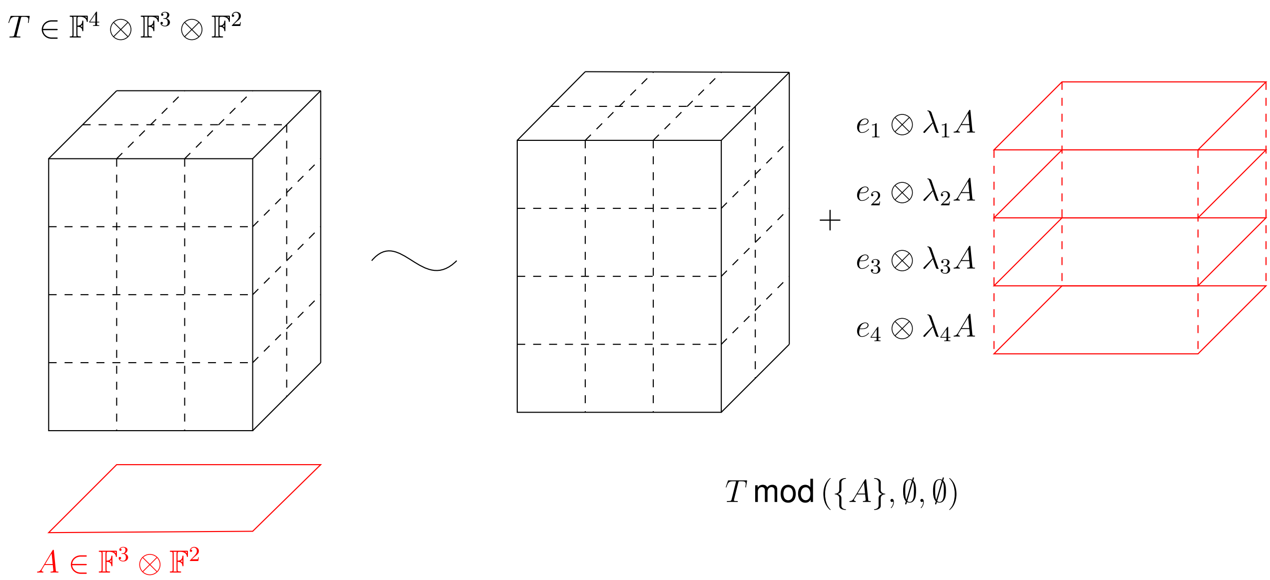

Figure 6: Family generated from a tensor T by a slice A

The above proof shows that the order of applying modulo operations on a given tensor does not affect the final resultant linear span.

Observation. Here, the family $F \subseteq V_1 \otimes \cdots \otimes V_d$, i.e. all its elements remain of the same order as the original tensor T. Figure 6 illustrates how one one such element from such a family can be generated.

Remark. As with the adjoining operation for symmetric tensors, generating a family of symmetric tensors modulo symmetric slices also follows immediately from the definitions provided above, but in this case it is the same family of order $d-1$ slices, $\mathcal{M}$, being used along every mode. Therefore T mod $\mathcal{M} := T$ mod $(\mathcal{M}, \ldots, \mathcal{M})$.

Since performing the modulo operation on a tensor gives a family instead of a single output tensor, one can talk about how the maximal or minimal rank of the tensor *family* is affected.

Combining Adjoining and Modulo

As seen in the previous two subsections, rank information of a tensor does change when adjoining slices, and the following theorem is becomes important in Wang and Seigal's 2023 paper [21].

Theorem 4.2.1 ([27]). *Suppose $T \in V := V_1 \otimes \cdots \otimes V_d$ is an order d tensor, and $\mathcal{M}_i \subset V_{\hat{i}}$ for every $1 \leq i \leq d$. Then*

$$\textbf{\textit{Rk}}(Adj(T, \mathcal{M}_1, \ldots, \mathcal{M}_d)) \geq \min \textbf{\textit{Rk}}(T \textit{ mod} (\mathcal{M}_1, \ldots, \mathcal{M}_d)) + \sum_{j=1}^{d} \dim\langle \mathcal{M}_j \rangle$$

Equality holds if each of the $\mathcal{M}_i$ consist of only rank-one tensors.

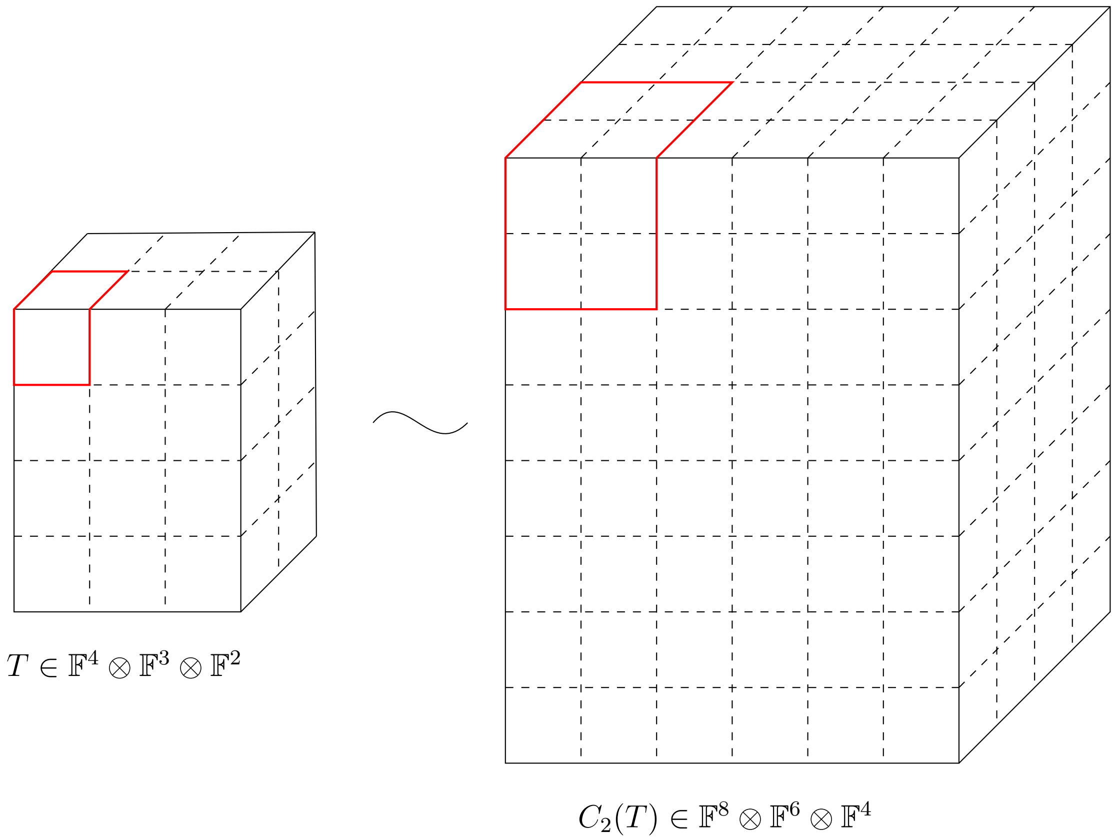

Figure 7: 2 clone of a tensor T

4.2.4 Cloning Tensors

The idea of cloning a tensor arises from necessity, because of difficulties that arise in the construction of a counterexample by Shitov, when constructing an emulating family. Cloning a tensor, as the name suggests, seeks to preserve the properties of the original, while embedding it in a much "larger space".

Definition 4.2.7 (Cloning a Tensor)**.** Fix $d > 0$. Let $T \in V := V_1 \otimes \cdots \otimes V_d$ be an order d-tensor.

The n-th cloning function $C_n : V \to (\bigoplus_{i=1}^n V_1) \otimes \cdots \otimes (\bigoplus_{i=1}^n V_d)$ can be defined in coordinates in the following manner:

$$C_n(T)(k_{1,j_1}, \ldots, k_{d,j_d}) = T(k_1, \ldots, k_d).$$

Here, the notation $k_{i,j_i} := k_{(j_i-1)n+i}$. where $1 \leq j_i \leq n$.

Observation. Since the definition of cloning depends on a non-zero d, one can clone tensors of any order, in particular vectors, as well. This leads to the description of the clone of a standard ordered basis vector $e_i \in V$, an $\mathbb{F}$ vector space as: $C_n(e_i) = \sum_{j=1}^n e_{i,j}$.

Lemma 4.2.2. *The cloning operation as defined above distributes over the tensor product i.e. $C_n(A \otimes B) = C_n(A) \otimes C_n(B)$ where A, B are vectors in an arbitrary vector space.*

Proof. Consider two arbitrary vectors, $A := (a_1, \dots, a_k), B := (b_1, \dots, b_k) \in V$, an $\mathbb{F}$ vector space. Then $A \otimes B$ is an order 2 tensor (a matrix) lying in $V \otimes V$, and $(A \otimes B)(i,j) = a_i b_j$ Then according to definition 4.2.7,

$$C_n(A \otimes B)(k_1, k_2) = (A \otimes B)(k_1(\bmod n), k_2(\bmod n)) = A(k_1(\bmod n))B(k_2(\bmod n)),$$

here using the regular modulo operation from number theory. This tensor lies in $(\bigoplus_{i=1}^n V) \otimes (\bigoplus_{i=1}^n V)$.

Now taking the vectors $C_n(A), C_n(B) \in (\bigoplus_{i=1}^n V)$. Here

$$C_n(A) = (\underbrace{a_1, \dots, a_1}_{n \text{ times}}, \dots \underbrace{a_k, \dots, a_k}_{n \text{ times}}),$$

and likewise for $C_n(B)$. Their tensor product is also a matrix lying in $(\bigoplus_{i=1}^n V) \otimes (\bigoplus_{i=1}^n V)$ where now

$$(C_n(A) \otimes C_n(B))(k_1, k_2) = C_n(A)(k_1)C_n(B)(k_2) = A(k_1(\bmod n))B(k_2(\bmod n)).$$

Thus, since each component of both tensors is equal, they are the same tensor. □

Another description of cloning is by first transforming the original by tensor product. Denote $\mathbf{1}_n$ as the vector of 1's lying in $\mathbb{F}^n$. Then the n-clone of a simple vector v can be described as $(v \otimes \mathbf{1}_n)^\emptyset$ or the flattening of the matrix created by tensoring v with the vector of ones. Thus another way to look at $C_n(T)$ is with the following procedure:

$$\begin{aligned} C_n(T) &= C_n\left(\sum_{i=1}^r T_i\right) \\ &= \sum_{i=1}^r C_n(T_i) \\ &= \sum_{i=1}^r C_n(v_1^i \otimes \cdots \otimes v_d^i) \\ &= \sum_{i=1}^r C_n(v_i^i) \otimes \cdots \otimes C_n(v_d^i) \\ &= \sum_{i=1}^r (v_1^i \otimes \mathbf{1}_n)^\emptyset \otimes \cdots \otimes (v_d^i \otimes \mathbf{1}_n)^\emptyset \end{aligned}$$

Remark. This property, along with the observation that cloning is a linear operation, allows for the proof that cloning a tensor does preserve its rank.

Proposition 4.2.2. ***Rk***(T) = ***Rk***$(C_n(T))$ *for* C_n *the cloning operation as defined in definition 4.2.7, and* $T \in V := V_1 \otimes \cdots \otimes V_d$.

Proof. The proof is straightforward, and is as follows

≤: Since T can be seen as a sub-tensor lying in $C_n(T)$, this inequality naturally exists. The rank of a subtensor is always at most the rank of the tensor, but never more.

≥: Since cloning distributes over addition (lemma 4.2.2), if one has the rank decomposition $T = \sum_{i=1}^r T_i$, then $C_n(T) = \sum_{i=1}^r C_n(T_i)$ is a decomposition of $C_n(T)$. Thus the rank of $C_n(T)$ is at most that of T.

□

4.3 The Counterexample over $\mathbb{R}$

Wang and Seigal [27] in 2023 published a paper that demonstrated the construction of a distinct counterexample to Comon's conjecture, following Shitov [23]. Their paper first examined the constructions made by Shitov, established new notation and terminology to prove results on rank pre- and post-operations, and finally they used MATLAB to construct a distinct order 6 example with real rank 30913 and real symmetric rank 30914.

As the title of the paper suggests, such a counterexample relies on certain lower bounds and inequalities remaining strict, particularly when making constructions in a field (namely $\mathbb{R}$) that is not algebraically complete.

4.3.1 Details from the Paper

The paper in 2023 [27] works over $\mathbb{R}$, specifically in $\mathbb{R}[x, y]$ as the underlying ring of polynomials from which the homogenous polynomials can be selected. For simplicity's sake, there are only 2 variables, corresponding to an underlying real vector space of dimension 2.

The particular tensor under consideration is represented by the homogenous polynomial $x^d - ky^d$ where $d > 0$ is a fixed, even integer, and $k > 0$ (also fixed). Such a homogenous polynomial corresponds to the following tensor

$$T := (e_1)^{\otimes d} - k(e_2)^{\otimes d}, k > 0, d = 2n, n \in \mathbb{Z}_+ \tag{4.1}$$

Visualised as a multiway array, this is a tensor that has non-zero entries at the two extreme "corners" and 0's everywhere else. Such a tensor clearly has rank and symmetric rank as 2.

Since the stated goal is to construct a counterexample to Comon's conjecture, this initial "seed" tensor, T, is first modified by the substitution method (section 4.2.3) to generate a linear span of tensors. Such a family when constructed has elements having minimal rank and symmetric rank. The elements in this family achieiving these minimal values do not have to be the same.

The family chosen in this case is

$$\mathcal{M} := \{x^{d-1-i}y^i\}_{i=1}^{d-2} \tag{4.2}$$

This is a collection of monomials of order $d-1$ that are of *mixed* form i.e. they are monomials not purely in either variable. When describing such a family in terms of symmetric tensors (see section 2.3), the family can be written as

$$\mathcal{M} := \left\{ \frac{1}{(d-1)!} \sum_{\sigma \in S_{d-1}} \sigma((e_1)^{\otimes d-1-i} \otimes (e_2)^{\otimes i}) \right\}_{i=1}^{d-2} \tag{4.3}$$

These order $d-1$ tensors, since not formed of purely only one standard ordered basis vector, do not affect the initial tensor T, when performing the operation of generating a family modulo slices: $T \bmod \mathcal{M}$. (Note here since they are working with symmetric tensors, it is enough to specify only one family.)

Establishing A First Difference in Rank and Symmetric Rank

Within the constructed family $T \bmod \mathcal{M}$, a tensor $P \in T \bmod \mathcal{M}$ corresponds to the following polynomial

$$P(x,y) := x^d - ky^d + \sum_{i=1}^{d-2}(\alpha_i x + \beta_i y)x^{d-1-i}y^i, \alpha_i, \beta_i \in \mathbb{R} \forall i = 1, \ldots, d-2, k > 0 \quad (4.4)$$

First, the symmetric rank decomposition is computed, and shown to be at least 2 or higher.

Proposition 4.3.1 ([27]). *A polynomial of the form $P(x,y)$ (as eq. (4.4)) has symmetric rank at least 2.*

Proof. Suppose not. Then one can rewrite $P(x,y)$ in terms of a linear polynomial raised to the dth power i.e.

$$P(x,y) = (ax+by)^d$$

for some choices of $a, b \in \mathbb{R}$.
Expanding the RHS would then lead to the following

$$\begin{aligned} P(x,y) &= (ax+by)^d \\ &= \sum_{j=0}^{d} \binom{d}{j} a^{d-j} b^j x^{d-j} y^j. \end{aligned}$$

This leads to the following equality of polynomial expressions:

$$x^d - ky^d + \sum_{i=1}^{d-2}(\alpha_i x + \beta_i y)x^{d-1-i}y^i = \sum_{j=0}^{d} \binom{d}{j} a^{d-j} b^j x^{d-j} y^j.$$

This leads to a contradiction simply by looking at the coefficient of the y^d monomial on either side of the equality. One must have $b^d = -k$ for some choice of $b \in \mathbb{R}$. Since such an equation has no real solutions, the initial assumption of being able to write P in terms of the d-th power of a single linear form was false, and therefore the Waring rank (equivalent to the symmetric rank) of $P \geq 2$. □

This establishes $\min_{P \in T \bmod \mathcal{M}} \mathbf{SRk}(P) = 2$. The next step in the paper is the establishing of the fact that the minimum *rank* element of such a family has rank 1.

Proposition 4.3.2 ([27]). *There is a tensor of the form $P(x,y)$ (see eq. (4.4)) that is an elementary tensor*

Proof. Consider the original tensor $T := x^d - ky^d$ (see eq. (4.1)). This is a tensor of the form eq. (4.4). Now summing all the elements in $\mathcal{M}$ as defined earlier (see eq. (4.2)) gives an order $d-1$ tensor with non-zero entries everywhere except for two "corners" — corresponding to x^{d-1} and y^{d-1}. These can be added along the d-th mode to the first slice, and after scaling by a factor of $-k$, to the last slice.

Formally the tensor T is now modified to

$$x^d - ky^d + \underbrace{\sum_{i=1}^{d-2} x^{d-i}y^i}_{\text{modifying the first } d\text{-slice}} + \underbrace{-k\left(\sum_{j=1}^{d-2} x^{d-1-j}y^{j+1}\right)}_{\text{scaling and modifying the last } d\text{-slice}}$$

An important point to note here is that changing the degree of the polynomial changes the order, and since the tensors all lie in $(\mathbb{R}^2)^{\otimes d}$, changes affecting the "first" slice have an extra factor of x, and changes affecting the "last" or second have an extra factor of y.

The final step of construction is adding the order-$(d-1)$ tensor corresponding to the monomial $x^{d-2}y$ (scaled by an appropriate a_i) to the first i-slice for all d modes of the tensor T. Likewise for the tensor corresponding to the monomial xy^{d-2} (with appropriate b_i added to the last i-slices). The choices of these scaling coefficients are such that the following conditions are satisfied:

$$\left(\sum_{i=1}^{d} a_i\right) - a_j = 0, \forall j \neq d \tag{4.5}$$

$$\sum_{i=1}^{d-1} a_i = -k \tag{4.6}$$

$$\left(\sum_{i=1}^{d} b_i\right) - b_j = 0, \forall j \neq d \tag{4.7}$$

$$\sum_{i=1}^{d-1} b_i = 1 \tag{4.8}$$

These choices for a_i are made because modifying the tensor by adding the scaled tensors modifies only one entry in last d slice. The last addition of $x^{d-1}y$ along the d-mode only affects the first d slice. A similar line of reasoning follows the choices for the b_i. Making such choices is always possible, and non-unique.

Performing these operations results in a tensor still lying in the family $T \bmod \mathcal{M}$, now of the form $(\mathbf{1}_2)^{\otimes d-1} \otimes (1, -k)$, an elementary order d tensor. Here again $\mathbf{1}_n$ denotes the vector of all 1's lying in $\mathbb{F}^n$ □

Remark. It is useful to note that proving such an elementary tensor exists is not a function of the underlying field. (assuming characteristic 0). Such a proof could also be modified for any number of variables (moving from working in $\mathbb{F}^2$ to $\mathbb{F}^n$), and for tensors of any even order d.

The following statement holds for the family $T \bmod \mathcal{M}$

$$\min \textbf{Rk}(T \bmod \mathcal{M}) < \min \textbf{SRk}(T \bmod \mathcal{M}) \tag{4.9}$$

Moving from Families to a Specific Tensor

Now that the existence of a family having elements with differing minimal rank and symmetric rank has been established, the authors construct $SAdj(T, \mathcal{M})$, as a symmetric tensor that is a candidate counterexample to Comon's conjecture. The problem of computing the rank of such a construction now arises, as it relates to the original T and candidate family $\mathcal{M}$.

The solution proposed originally by Shitov [23], is a proposed modification of $\mathcal{M}$ to a family of *elementary* order $(d-1)$ tensors that span the same linear space. Unfortunately, in the format $(\mathbb{R}^2)^{\otimes 4}$, doing so also violates the vital inequality required for rank and symmetric rank of the family $T \bmod \mathcal{M}$ (see eq. (4.9))

Such a modification is required for the original computation of rank (see theorem 4.2.1) to be an equality (because equality only holds when the family consists of elementary tensors).

The following proposition is adapted from [27], to show that for higher order examples modifying $\mathcal{M}$ in such a direct way as above does not violate eq. (4.9)..

Proposition 4.3.3. *Consider T as in eq. (4.1), and $\mathcal{M}$ as eq. (4.2). Let $\mathcal{W} \subset (\mathbb{R}^2)^{\otimes d-1}$, a finite set of elementary symmetric tensors such that $\mathcal{W}$ spans $\mathcal{M}$. Then $0 \notin T \bmod \mathcal{W}$ when $d = 2n, n > 2$.*

Proof. Consider $\mathcal{W}$, a collection of elementary symmetric tensors of appropriate order. Showing $0 \in T \bmod \mathcal{W}$ is equivalent to asking that $\mathcal{W}$ spans the set $\{x^i y^d - i\}_{i=0}^{d-1} \supseteq \mathcal{M}$ This is due to the fact that every slice of T (seen as an order $d-1$ tensor and therefore represented by a degree $d-1$ polynomial) can then be expressed as lying in the span of $\mathcal{W}$. In particular, if one can show that the dimension of span $\mathcal{W}$ is d (the number of degree $d-1$ monomials in 2 variables), this completes the proof.

Suppose the dimension of the span of $\mathcal{W}$ is less than d. Then consider the tensor represented by $x^{d-2}y$. This monomial has rank $d-1$ since it can be written minimally as sum of $d-1$ elementary tensors. Thus the dimension of span $\mathcal{W}$ is $d-1$. Since $\mathcal{W}$ consists of symmetric tensors, its basis vectors must be a linear forms raised to the $d-1$th power: $(\lambda_i x + \mu_i y)^{d-1}$. These forms must also correspond to elementary tensors, by virtue of being elements of $\mathcal{W}$.

Now since this set, $\{(\lambda_i x + \mu_i y)^{d-1}\}_{i=1}^{d-1}$ is a basis for span $\mathcal{W}$, one must have that the tensors $x^{d-2}y$ and xy^{d-2} can be written as linear combinations of the elements of the set. Doing so imposes conditions on $\lambda_i, \mu_i > 0$ such that

$$\sum_{i=1}^{d} \frac{\lambda_i}{\mu_i} = 0 = \sum_{i=1}^{d} \frac{\mu_i}{\lambda_i}.$$

This set of equalities results in

$$1 = \frac{\lambda_1}{\mu_1}\frac{\mu_1}{\lambda_1} = \left(\sum_{i=2}^{d} \frac{\lambda_i}{\mu_i}\right)\left(\sum_{i=2}^{d} \frac{\mu_i}{\lambda_i}\right) = (d-1) + \sum_{i=2}^{d}\sum_{j\neq i} \frac{\lambda_i \mu_j}{\mu_i \lambda_j}.$$

If one can show that $\sum_{i=2}^{d}\sum_{j<i} t_{ij} + t_{ij}^{-1} + d - 2 \neq 0$, where $t_{ij} := \frac{\lambda_i \mu_j}{\mu_i \lambda_j}$, then the initial assumption of span $\mathcal{W}$ being of a dimension less than d is false, since this leads to a contradiction. However, the range of this function is the entirety of $\mathbb{R}$ whenever dealing with a value of $d > 4$ since the polynomial generated is no longer univariate. Therefore the initial goal of showing $0 \in T \bmod \mathcal{W}$ was false.

$\square$

4.3.2 Emulating Family Construction

While for the order 4 case from Shitov, the issue of eq. (4.9) being violated when modifying the family $\mathcal{M}$ arose, as seen in the previous proposition, such issues do not arise when working with the order 6 or higher cases. However, the authors of [27] continue to use the cloning methodology later in their paper, and doing so ensures that there are no further complications that could arise when modifying the initial family of slices $\mathcal{M}$.

As demonstrated in section 4.2.4, since cloning preserves rank, then inequality is preserved after taking the k-th clones, and transforms into

$$\min \textbf{Rk} C_k(T) \operatorname{mod} C_k(\mathcal{M}) < \min \textbf{SRk} C_k(T) \operatorname{mod} C_k(\mathcal{M}) \tag{4.10}$$

In this situation, modifying $C_k(M)$ (a family of order $d-1$ tensors lying in $C_k(V^{\otimes d-1})$) into $\mathcal{W}$ such that the above modification procedure can occur successfully. This results in working with a symmetric tensor $SAdj(C_k(T), \mathcal{W})$ - the final symmetric tensor having a definite distinction between rank and symmetric rank.

It is past this point that both Shitov [23], and Wang and Seigal [27] go forward with the construction of a family $\mathcal{W}$ to replace $C_k(\mathcal{M})$ that can be described algorithmically, but the justification for why such choices are made remains opaque.

The later paper [27] constructs this family for their candidate counterexample of order 6 as described below. In personal correspondence, the lead author indicated that this falls in step with the construction in the preprint [23]:

They begin with vectors lying in $\mathbb{R}^{2n}$ of the form a_i, where $i \in [n]$ and each vector is such that $a_i(2i) = a_i(2i-1) = 1$ and all the other entries are 0. Then the new vectors constructed lie in $\mathbb{R}^{4n}$ and are one of the following forms (where the $\mid$ indicates the distinction between the first $2n$ entries and the last $2n$ entries.)

1. $(a_{i_1} + a_{i_2} + a_{i_3} + a_{i_4} \mid a_{k_1})$
2. $(a_{i_1} + a_{i_2} + a_{i_3} + a_{i_4} \mid 0)$
3. $(a_{i_1} + a_{i_2} + a_{i_3} \mid \frac{n-2}{n-1} a_{k_1})$
4. $(a_{i_1} + a_{i_2} + a_{i_3} \mid \frac{n-3}{n-4} a_{k_1})$
5. $(a_{i_1} + a_{i_2} + a_{i_3} \mid a_{k_1} + a_{k_2})$
6. $(a_{i_1} + a_{i_2} + a_{i_3} \mid 0)$
7. $(a_{i_1} + a_{i_2} \mid \frac{(n-2)^2}{(n-3)(n-1)} a_{k_1})$
8. $(a_{i_1} + a_{i_2} \mid \frac{n-2}{n-4} a_{k_1})$
9. $(a_{i_1} + a_{i_2} \mid \frac{n-2}{n-3} (a_{k_1} + a_{k_2}))$
10. $(a_{i_1} + a_{i_2} \mid 0)$
11. $(a_{i_1} \mid \frac{n-2}{n-3} a_{k_1})$
12. $(a_{i_1} \mid \frac{n-1}{n-4} a_{k_1})$

13. $(a_{i_1} \mid \frac{n-1}{n-3}(a_{k_1} + a_{k_2}))$

14. $(a_{i_1} \mid 0)$

15. $(0 \mid a_{k_1})$

16. $(0 \mid a_{k_1} + a_{k_2})$

Here $1 \leq i_1 < i_2 < i_3 < i_4 \leq n$ and $1 \leq k_1 < k_2 \leq n$. The collection of all the 16 vectors in $\mathbb{R}^{4n}$ described above, for all choices of indices i_1, i_2, i_3, i_4 and k_1, k_2 forms the set $\mathcal{W}_1$. This is only half of their construction.

Under the permutation operation

$$\pi : (i_1, i_2, \ldots, i_{2n}, k_1, k_2, \ldots, k_{2n}) \mapsto (k_2, \ldots, k_{2n}, k_1, i_2, \ldots, i_{2n}, i_1),$$

the same set $\mathcal{W}_1$ can be transformed into a new set $\mathcal{W}_2$. The final emulating family they use for their order 6 counterexample is $\mathcal{W} := \mathcal{W}_1 \cup \mathcal{W}_2$

This emulating family proves to be the last piece in their construction of $SAdj(C_k(T), \mathcal{W})$, an order 6 symmetric tensor having real symmetric rank 30914 and real rank 30913.

Extending this counterexample to higher orders requires simply extending these choices in an appropriate manner, or a more clear understanding of the choices made in the construction of $\mathcal{W}$.

Chapter 5

Conclusion

While this thesis only condenses the literature surrounding Comon's conjecture and attempts to generate counterexamples to it, the fact that a computed, real counterexample does exist provides some hope for a conclusive answer in the future. As mentioned in chapter 4, since the real counterexample requires the discrepancy eq. (4.9), an equivalent condition in the complex case seems to be the pathway forward.

To be more precise: if a symmetric, complex-valued, order d tensor T exists, along with a family of symmetric, complex-valued, order $d-1$ tensors $\mathcal{M}$, such that the statement

$$\min \mathbf{Rk} T \operatorname{mod} \mathcal{M} < \min \mathbf{SRk} T \operatorname{mod} \mathcal{M}$$

still holds, then a viable candidate counterexample can be constructed, paralleling the constructions in [23, 27], since the only part of these constructions dependent on the underlying field is the choice of candidate real $T_{\mathbb{R}}$ represented by the homogenous even order d polynomial $x^d - ky^d$.

5.1 Further Possible Work

The work in this thesis establishes a baseline for exploring constructions of candidate counterexamples to Comon's Conjecture.

- One way this work can be extended is to focus more on a geometric description of the constructions made in this paper, particularly for eq. (4.9). In doing so, the hope is that one would uncover geometric conditions that indicate a candidate counterexample to the conjecture, and these are hopefully independent of the base field chosen.

- A more explicit understanding of the choices of vectors made in section 4.3.2. The choices made so far stem from [23], and while they have proved to be effective, have no indication as to why they are. Studying the construction of such a family more carefully might indicate algebraic or geometric conditions that are being met by the collection of slices, allowing for the existence of an emulating family of elementary tensors.

- Comon's conjecture as originally stated in [19] was only for $\mathbb{R}$- or $\mathbb{C}$-valued tensors. A conclusive understanding of whether Comon's conjecture can be extended to finite fields, or if such an extension is non-trivial can be one path forward.

- For the real, even order case, the developement of explicit algorithms that compute the optimal rank decomposition for a given symmetric decomposition is another potential avenue of work. The guiding principle is of the Gram-Schmidt process that allows for any collection of n vectors to be made into an orthonormal basis for an n dimensional vector space.

- Adjusting the work in this paper to understand the smallest possible cloning operation required to satisfy the conditions of the emulating family could be another potential extension of this work. The paper from Borovik et a. in 2025 [5], while not explicitly related to Comon's conjecture, does establish an upper bound for how large a cloning operation may be required.

- Since the relationship between symmetric tensors and homogenous polynomials is explicit, extending that mapping to show a general relationship between tensors having a certain order, and polynomials having the same degree. The paper from Comon et al in 2008 [13] indicates that such an extension may not be possible directly.

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Department of Mathematics
Celestijnenlaan 200B
3001 LEUVEN (HEVERLEE), BELGIË
tel. + 32 16 32 70 07
fax + 32 16 32 79 98
wis.kuleuven.be